Detailed Runtime & Performance Breakdown

Comprehensive timing across components. epyKit remains fastest by a wide margin.

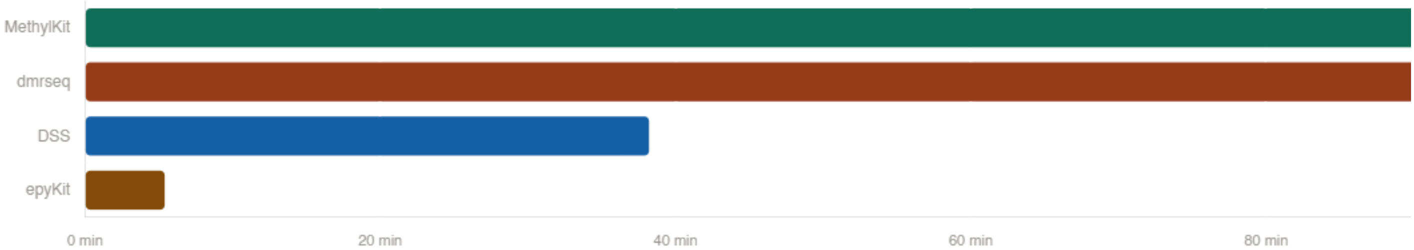
epyKit **fastest**
5.4 min
324 s wall · 8 cores

DSS
38.2 min
2,292 s wall · 2 cores

dmrseq
130.6 min
7,837 s wall

MethylKit
148.7 min
8,920 s wall · 2 cores

epyKit DSS dmrseq MethylKit



epyKit

Ingest	253 s
DMC test (lr+eb)	58 s
DMR chain merge	13 s
Total	**324 s**

DSS

Ingest	54 s
DML test + smooth	2,234 s
callDMR	4 s
Total	**2,292 s**

dmrseq

Ingest & filter	78 s
dmrseq calling	7,739 s
Annotation	20 s
Total	**7,837 s**

MethylKit

Ingest	74 s
DMC test (Chisq)	8,019 s
DMR tiling	809 s
Total	**8,920 s**

Run completed 2026-06-22 · 4 tools · 5h 19m total · reference: 813 paper DMRs (AKAP11-KO vs WT)

DSS sig DMRs
24,786
runtime 39 min

MethylKit sig DMRs
26,166
runtime 2 h 29 min

epyKit sig DMRs
1,139
not in pipeline run

dmrseq sig DMRs
5
exit code 1 · 130.6 min

epyKit sig DMCs
48,184
single-CpG level

MethylKit sig DMCs
75,790
single-CpG level

Sensitivity vs. reference paper

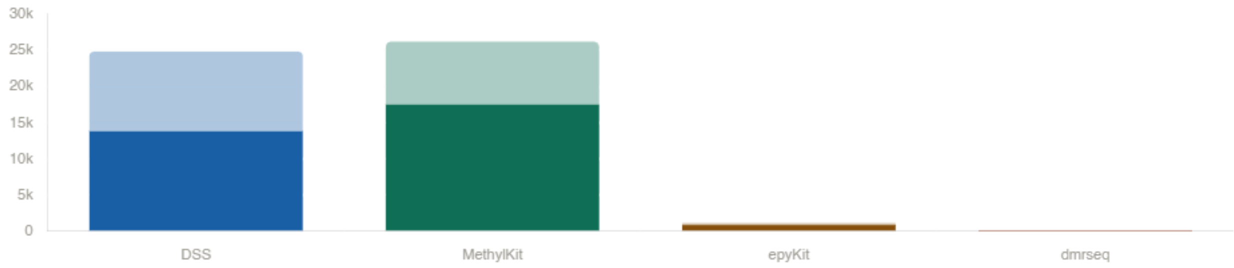
Fraction of the 813 published DMRs recovered by each tool (genomic overlap)

DSS MethylKit epyKit dmrseq (sig) dmrseq (all)

DSS recovers 797/813 (98.0%) of paper DMRs with the fastest runtime. **MethylKit** and **epyKit** are less exhaustive (79.8% / 74.7%) but call far fewer false-positive-like regions overall. **dmrseq** exits with a non-zero return code and yields only 5 significant regions, recovering a mere 1.6% — but its full (unsignificant-filtered) output covers 95.2%, suggesting the issue is the internal significance threshold, not the underlying model.

DMR counts & hyper/hypo balance

Absolute count and directional split for each caller

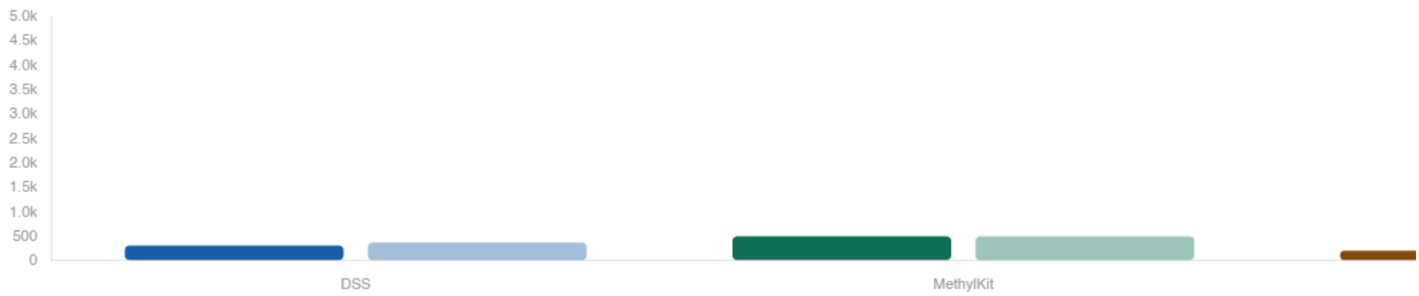


Tool	Total DMRs	Hyper	Hypo	Hyper %
DSS	24,786	13,803	10,983	55.7%
MethylKit	26,166	17,481	8,685	66.8%
epyKit	1,139	859	280	75.4%
dmrseq (sig)	5	5	0	100%

All tools agree on a **hypermethylation bias** in AKAP11-KO clones relative to WT, consistent with the original paper. epyKit's bias is strongest (75% hyper). dmrseq's 5-region output is 100% hyper, but the tiny N makes this uninformative.

DMR size distribution

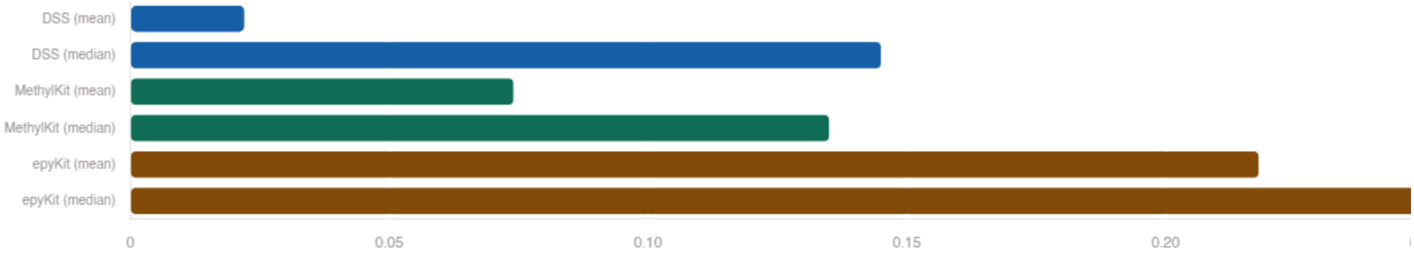
Median and mean genomic span (bp) of called DMRs



Tool	Median bp	Mean bp	Max bp	Window strategy
DSS	310	372	7,611	adaptive
MethylKit	499	499	499	fixed 500 bp tiles
epyKit	205	235	2,560	chain-merge Δ250
dmrseq (sig)	4,869	3,630	5,794	model-based

Methylation difference magnitude

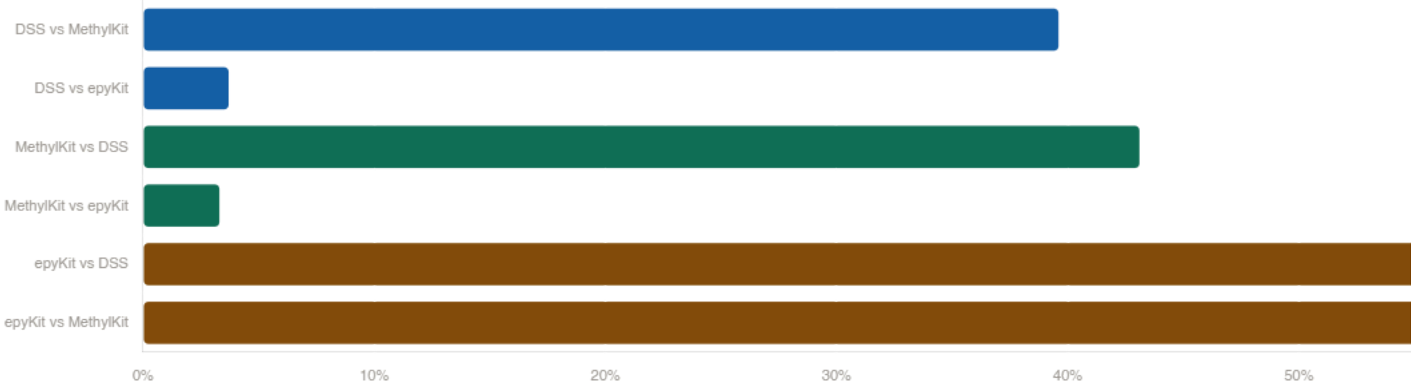
Mean absolute methylation difference reported by each tool (DMR level)



Tool	Mean |Δmeth|	Median	Scale
DSS	0.022	0.145	0–1 (proportion)
MethylKit	7.4%	13.5%	0–100 (%)
epyKit	0.218	0.382	0–1 (proportion)
dmrseq (sig)	—	—	area statistic

Tool–tool overlap (DMR level)

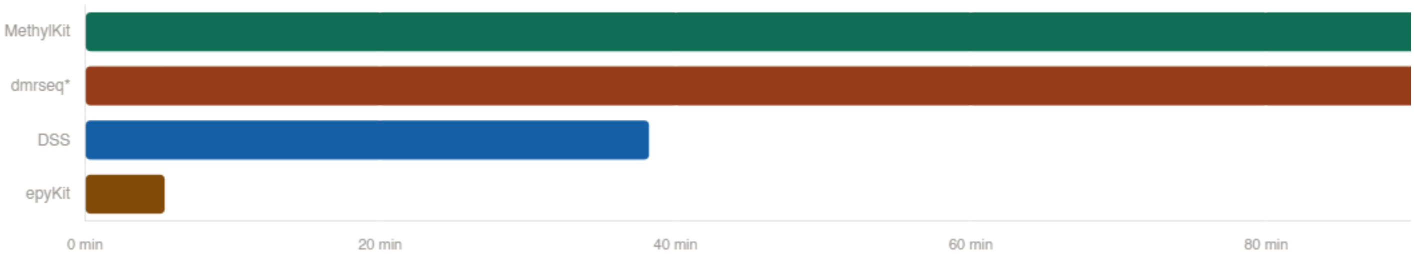
Fraction of each tool's DMRs that overlap ≥1 region from another caller



Source	vs DSS	vs MethylKit	vs epyKit
DSS (n=24,786)	—	39.6%	3.7%
MethylKit (n=26,166)	43.1%	—	3.3%
epyKit (n=1,139)	84.0%	62.6%	—

Runtime & pipeline status

From runner_summary_20260622_170821.json and newly provided JSON



Tool	Runtime	Exit code	Status
DSS	38 min 53 s	0	OK
MethylKit	2 h 29 min	0	OK
dmrseq	130.6 min	1	Error

Summary scorecard

Qualitative assessment across key dimensions

Dimension	DSS	MethylKit	epyKit	dmrseq
Paper sensitivity	★★★ 98%	★★ 80%	★★ 75%	★ 2% (sig)
Precision / specificity	★ very broad	★★ broad	★★★ narrow	n/a
Effect size magnitude	★★ mixed	★★ mixed	★★★ high	n/a
DMR resolution	★★ 310 bp median	★ fixed 500 bp	★★★ 205 bp	★ large regions
Speed	★★★ 39 min	★ 149 min	★★ (not in run)	★ 131 min + error
Hyper/hypo balance	★★★ 56/44	★★ 67/33	★ 75/25	100/0 (n=5)
Pipeline stability	★★★	★★★	★★★	★ exit code 1